

Carnage

Aim:

To analyze a malicious packet capture (PCAP) using Wireshark and identify infection behavior, command-and-control (C2) communication, and post-infection activity.

1. Scenario

Eric Fischer from the *Purchasing Department* at *Bartell Ltd* has received an email from a known contact with a Word document attachment. Upon opening the document, he accidentally clicked on “Enable Content.” The *SOC Department* immediately received an alert from the endpoint agent that Eric’s workstation was making suspicious connections outbound. The *pcap* was retrieved from the network sensor and handed to you for analysis.

Task: Investigate the packet capture and uncover the malicious activities.

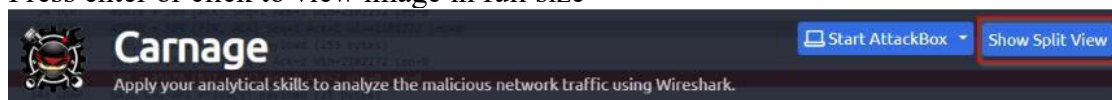
*Credit goes to [Brad Duncan](#) for capturing the traffic and sharing the pcap packet capture with InfoSec community.

NOTE: DO NOT directly interact with any domains and IP addresses in this challenge.

Deploy the machine attached to this task; it will be visible in the **split-screen** view once it is ready.

If you don’t see a virtual machine load, then click the **Show Split View** button.

Press enter or click to view image in full size



This image and all the theoretical content of the present article is TryHackMe’s property.

Answer the question below

1.1. Read the above.

No answer needed

2. Traffic Analysis

Press enter or click to view image in full size



This image and all the theoretical content of the present article is TryHackMe's property.

Are you ready for the journey?

Please, load the *pcap* file in your *Analysis* folder on the Desktop into *Wireshark* to answer the questions below.

Answer the questions below

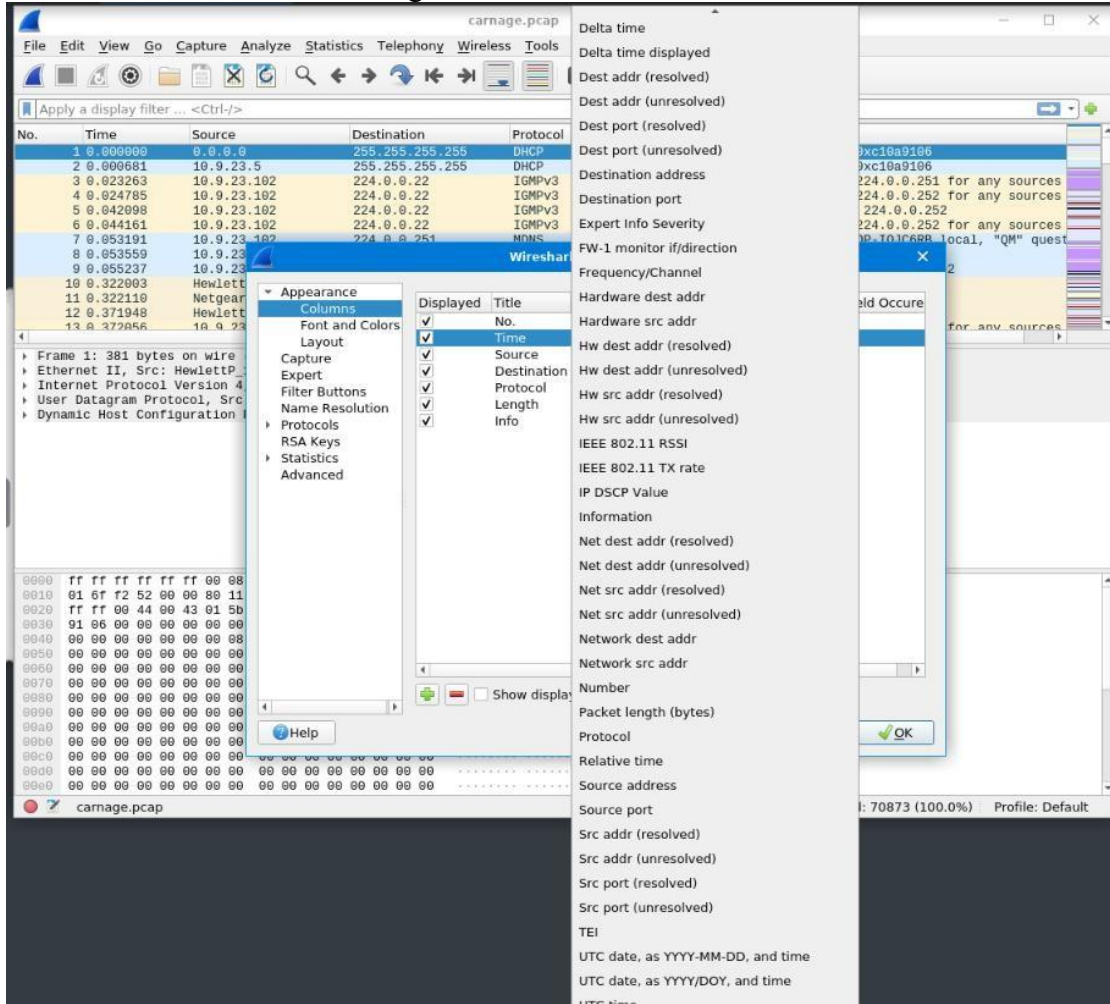
2.1. What was the **date and time** for the **first HTTP connection** to the malicious IP? (**answer format**: yyyy-mm-dd hh:mm:ss)

2021-09-24 16:44:38

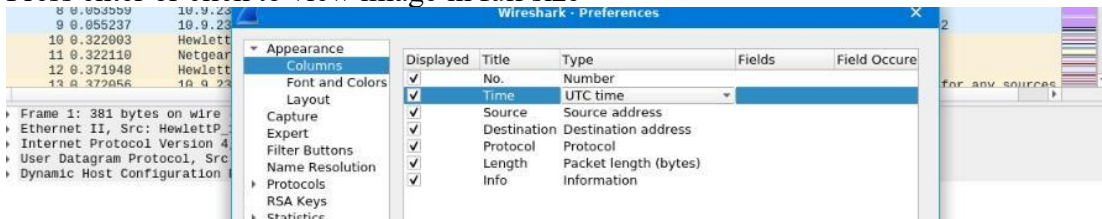
Configured date and time format.

Clicked *Edit > Preferences > Columns > Time > UTC date, as YYYY-MM-DD, and time > OK.*

Press enter or click to view image in full size

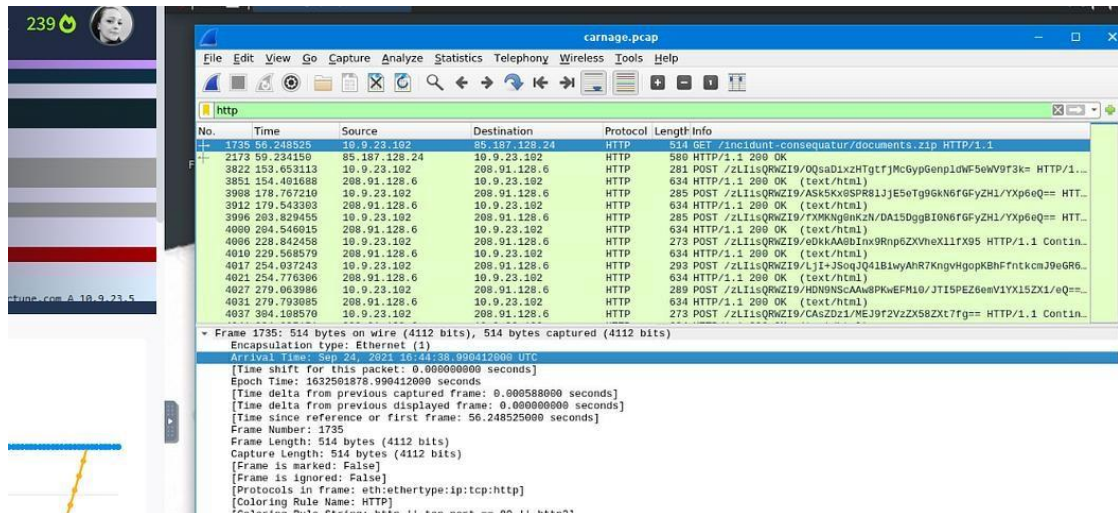


Press enter or click to view image in full size



Applied filter http.

Press enter or click to view image in full size

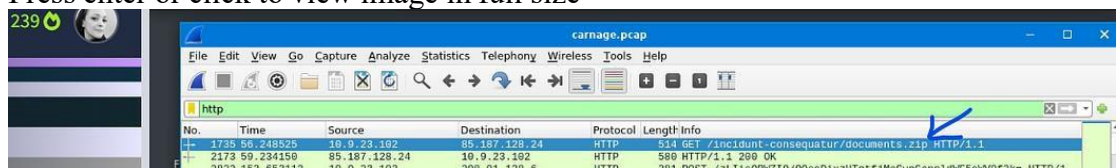


2.2. What is the name of the zip file that was downloaded?

documents.zip

Got the answer in the frame analyzed in 2.1.

Press enter or click to view image in full size

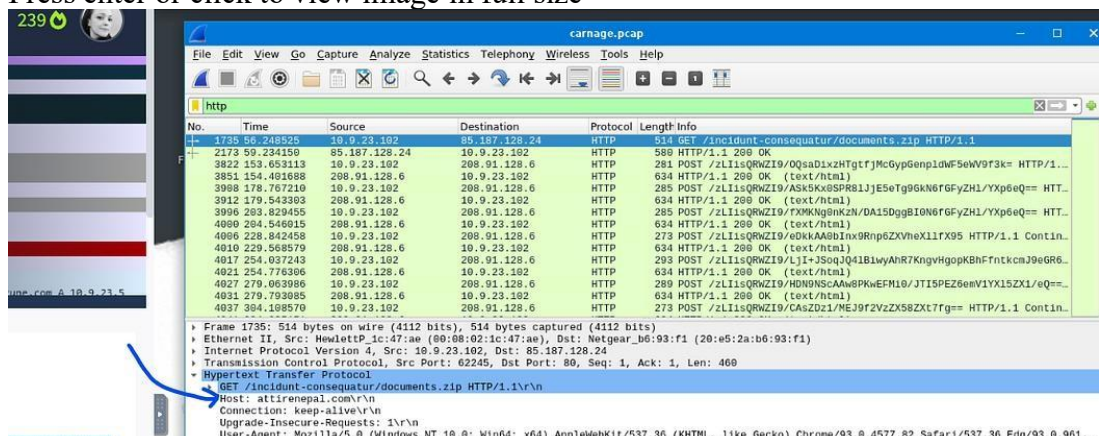


2.3. What was the domain hosting the malicious zip file?

attirenepal.com

Got the answer in the frame analyzed in 2.1: clicked over the frame and then over *Hyper Text Transfer Protocol*.

Press enter or click to view image in full size

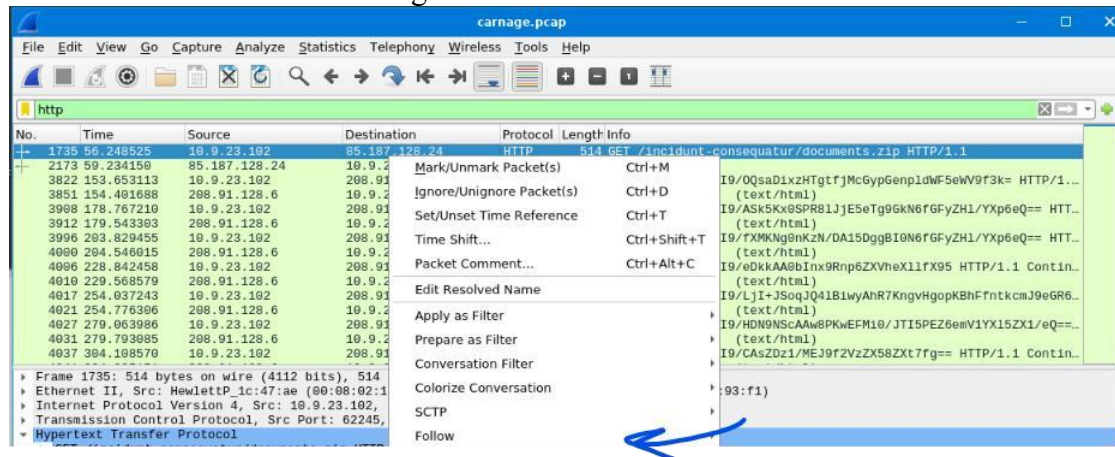


2.4. Without downloading the file, what is the name of the file in the zip file?

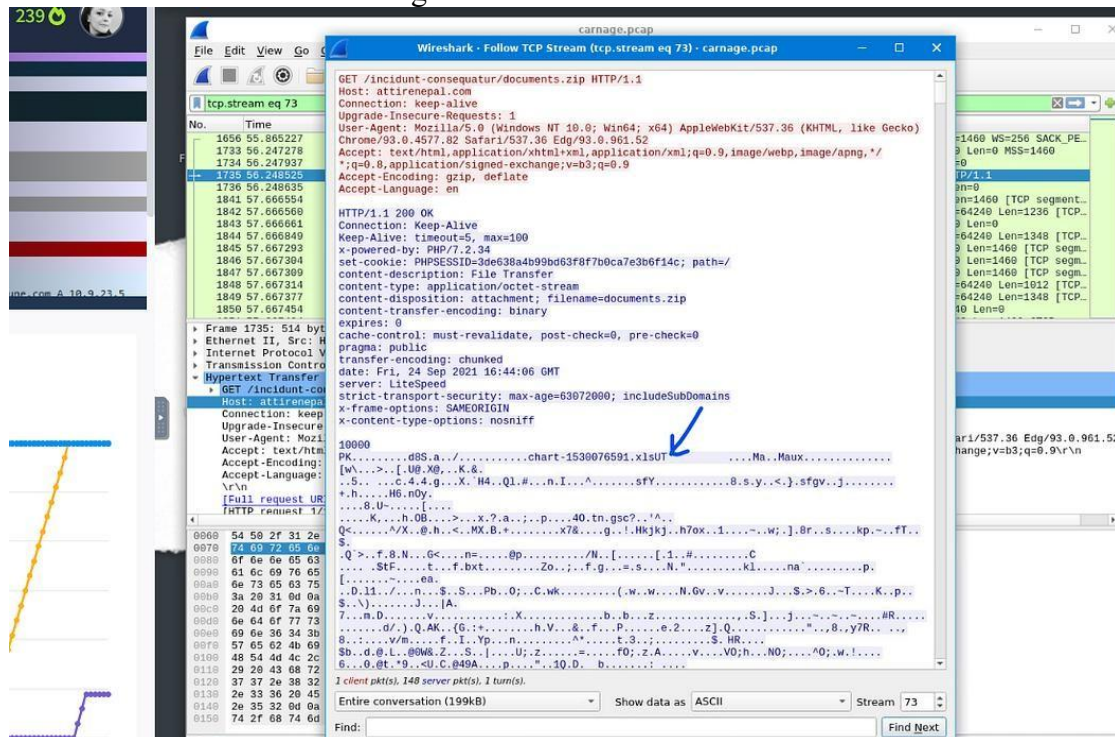
chart-1530076591.txt

Got the answer in the frame analyzed in 2.1: right-clicked over the frame, selected *Follow*, selected *TCP Stream*.

Press enter or click to view image in full size



Press enter or click to view image in full size

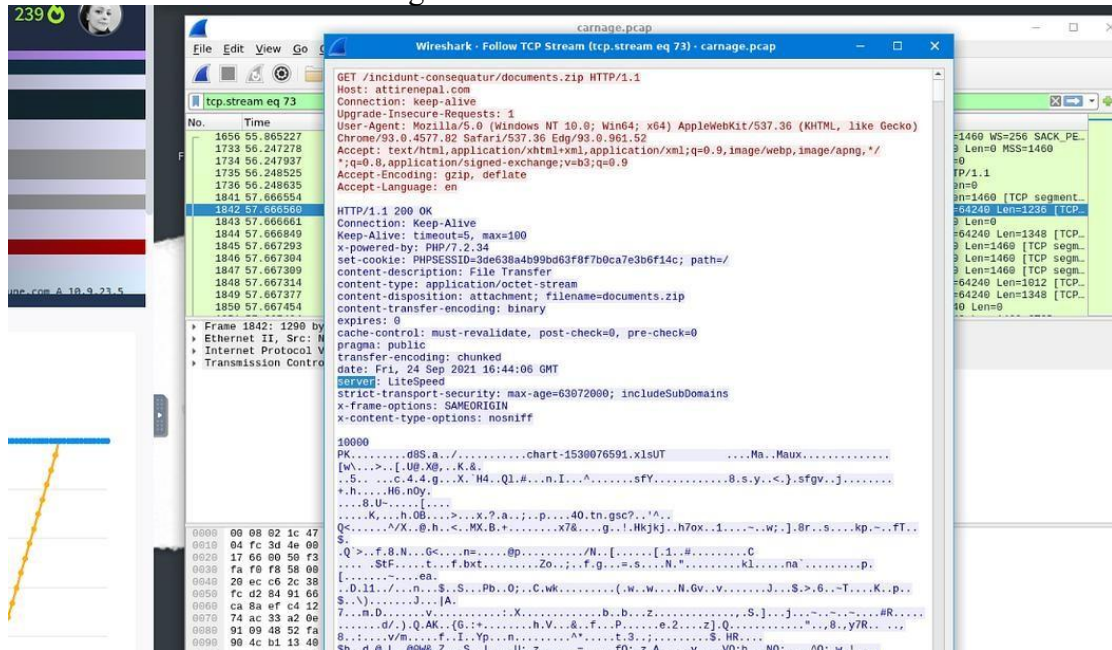


2.5. What is the name of the webserver of the malicious IP from which the zip file was downloaded?

LiteSpeed

Got the answer in the screen we used in 2.4.

Press enter or click to view image in full size

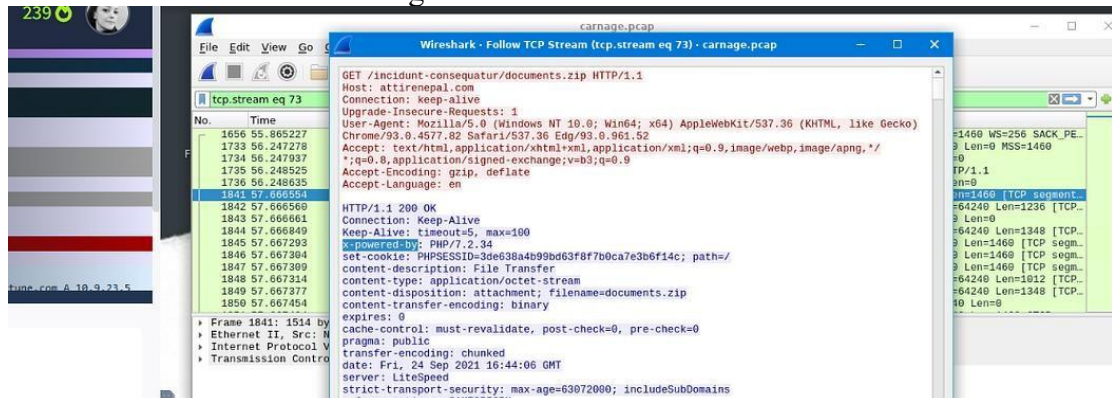


2.6. What is the **version of the webserver** from the previous question?

1.6. PHP/7.2.34

Got the answer in the screen we used in 2.4.

Press enter or click to view image in full size



2.7. **Malicious files** were downloaded to the victim host from multiple domains.

What were the **three domains involved** with this activity?

finejewels.com, thietbiagt.com, new.americold.com

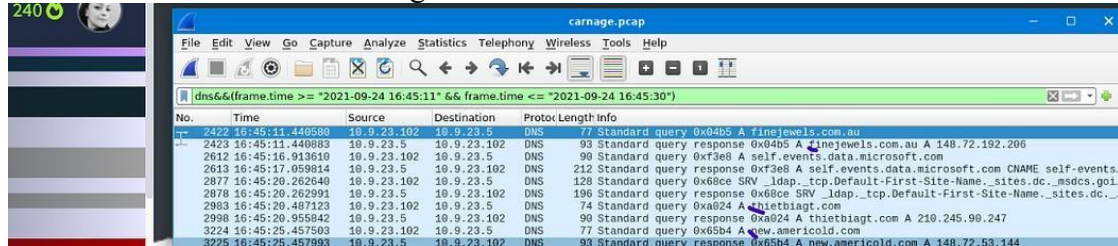
Cleared the previous filter. Applied *dns* as display filter.

The room provides a hint: check HTTPS traffic. Narrow down the timeframe from 16:45:11 to 16:45:30.

Updated the filter as shown below

`dns && (frame.time >= "2021-09-24 16:45:11" && frame.time <= "2021-09-24 16:45:30")`

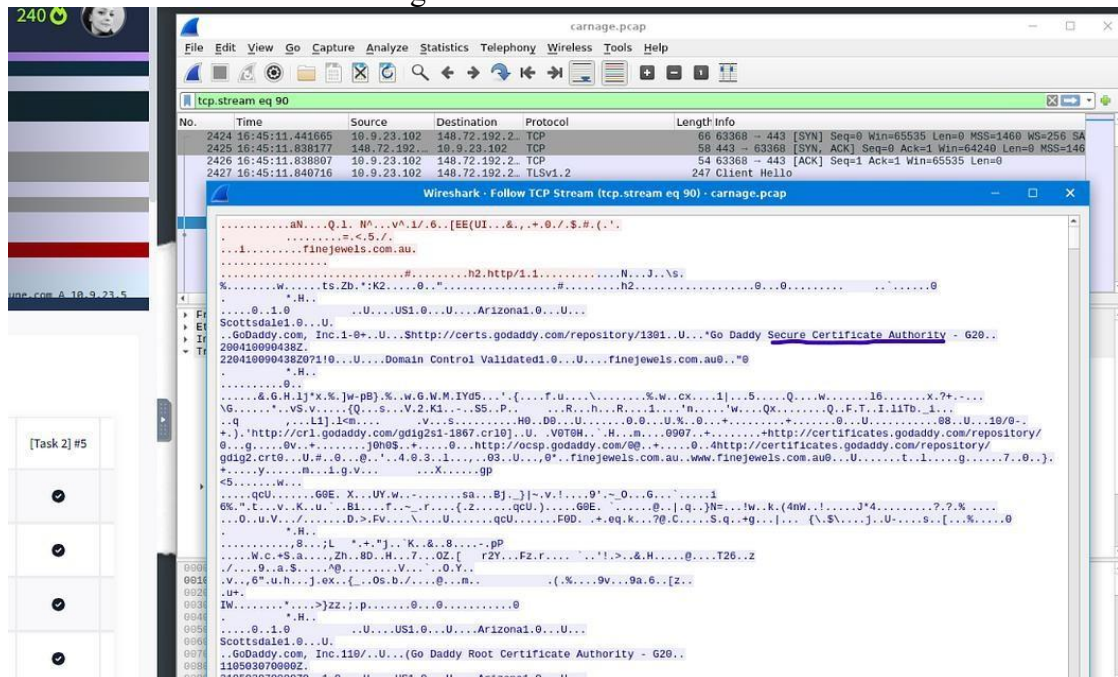
Press enter or click to view image in full size



2.8. Which certificate authority issued the SSL certificate to the first domain from the previous question?

GoDaddy

Press enter or click to view image in full size



2.9. What are the two IP addresses of the Cobalt Strike servers? Use VirusTotal (the Community tab) to confirm if IPs are identified as Cobalt Strike

C2 servers. (answer format: enter the IP addresses in sequential order)

185.106.96.158, 185.125.204.174

In the main menu bar clicked *Statistics*, then *Conversations*. Chose *TCP*.

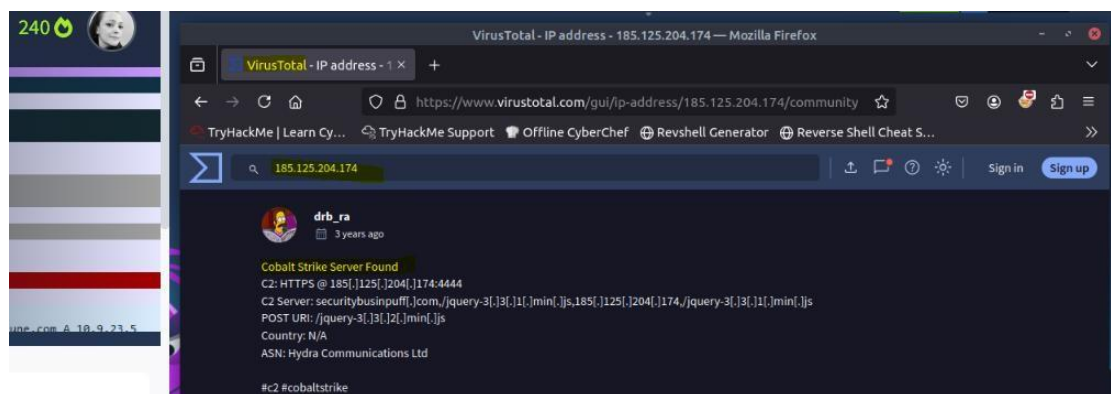
Ordered results by column *Packets*.

Press enter or click to view image in full size

Address A	Port A	Address B	Port B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start
23.111.114.52	65400	10.9.23.102	63557	18002	15M	12281	15M	5721	753k	884.211305
10.9.23.102	63555	104.83.84.137	443	9074	10M	1972	107k	7102	10M	880.206501
10.9.23.102	63465	185.125.204.174	8080	1375	1387k	408	22k	967	1364k	743.649672
10.9.23.102	63507	185.106.96.158	80	1074	997k	379	20k	695	976k	799.905706
10.9.23.102	63723	177.149.159.181	25	1069	685k	464	652k	605	33k	1180.52695
10.9.23.102	63439	136.232.34.70	443	1002	989k	284	16k	718	973k	647.865515
10.9.23.102	63726	52.97.201.242	25	978	627k	446	592k	532	34k	1186.87597
10.9.23.102	63610	136.232.34.70	443	976	882k	334	19k	642	862k	1057.18350
10.9.23.102	63732	52.97.201.210	25	957	616k	441	583k	516	33k	1211.75241
10.9.23.102	63571	136.232.34.70	443	953	910k	291	16k	662	893k	932.806205
10.9.23.102	63736	52.97.201.242	25	945	596k	416	562k	529	34k	1219.20120
10.9.23.102	63738	217.70.178.9	25	933	585k	405	551k	528	34k	1224.08126
10.9.23.102	63734	62.149.128.200	25	930	574k	397	540k	533	34k	1214.06720
10.9.23.102	63747	52.98.163.18	25	922	579k	410	543k	512	35k	1241.90272
10.9.23.102	63752	185.14.56.240	25	900	573k	392	545k	508	27k	1258.19607
10.9.23.102	63744	52.97.201.194	25	898	574k	402	541k	496	32k	1232.91586
10.9.23.102	63694	52.97.201.242	25	886	577k	415	544k	471	33k	1145.30251
10.9.23.102	63749	45.64.187.254	25	883	573k	399	542k	484	31k	1248.38420
10.9.23.102	63740	52.98.168.178	25	882	585k	412	554k	470	30k	1226.38003
10.9.23.102	63725	52.97.201.242	25	878	572k	411	541k	467	30k	1186.16735
10.9.23.102	63737	212.227.15.158	25	877	579k	410	549k	467	29k	1219.92574
10.9.23.102	63745	193.70.18.144	25	866	581k	416	551k	450	30k	1240.69625
10.9.23.102	63729	52.97.232.210	25	862	573k	401	543k	461	30k	1201.59966
10.9.23.102	63730	46.16.61.50	25	861	569k	405	539k	456	30k	1208.37763
10.9.23.102	63707	154.41.66.35	25	860	587k	423	558k	437	28k	1152.44779
10.9.23.102	63741	193.70.18.144	25	860	581k	403	550k	457	30k	1229.63649

Researched in *VirusTotal* and confirmed that it is about *Cobalt Strike*.

Press enter or click to view image in full size



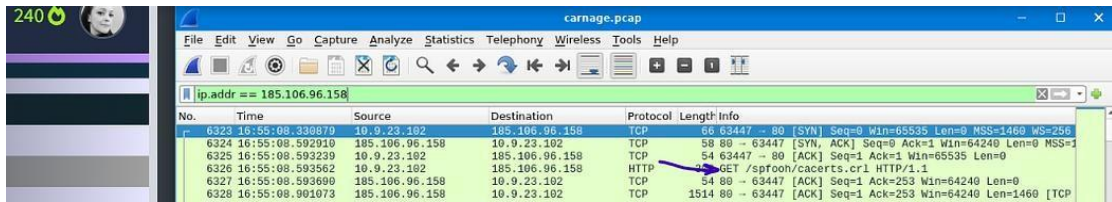
2.10. What is the **Host** header for the **first Cobalt Strike** IP address from the previous question?

ocsp.verisign.com

Used filter `ip.addr == 185.106.96.158`

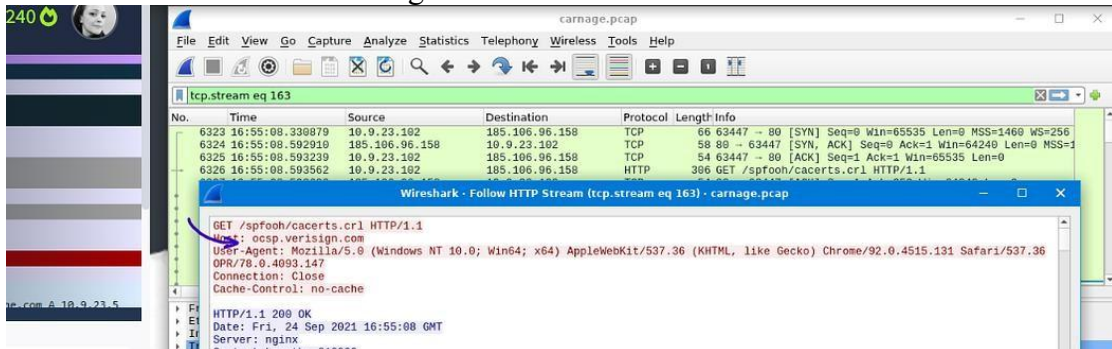
Searched for *GET*.

Press enter or click to view image in full size



Followed its HTTP Stream.

Press enter or click to view image in full size



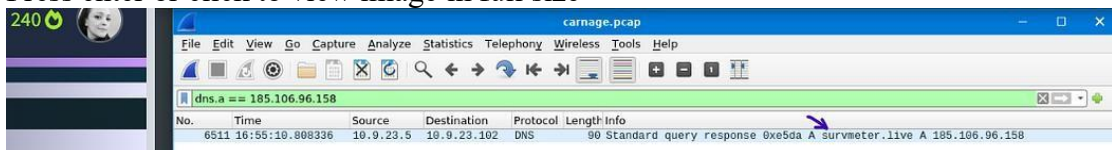
2.11. What is the **domain name** for the **first IP address of the Cobalt Strike server**? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

survmeter.live

It was provided a hint: *Filter out for dns queries.*

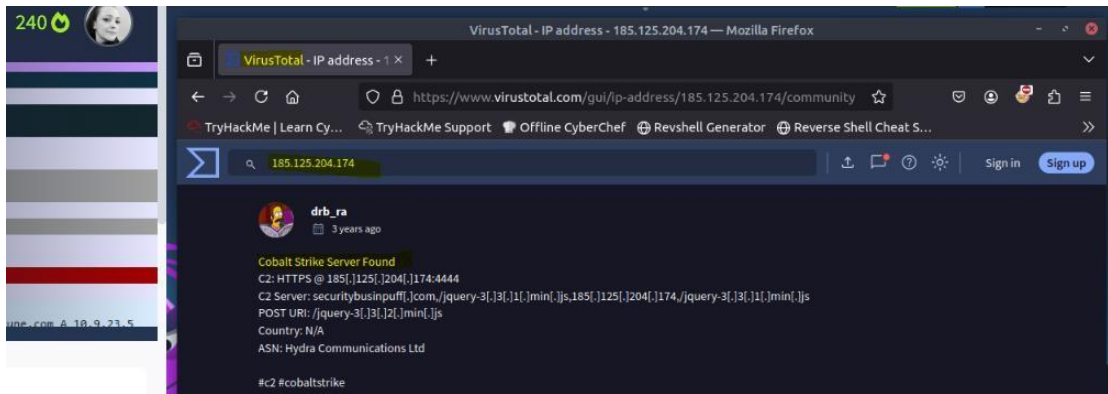
Applied filter: `dns.a == 185.106.96.158`

Press enter or click to view image in full size



Could also read the information provided in *VirtusTotal*.

Press enter or click to view image in full size

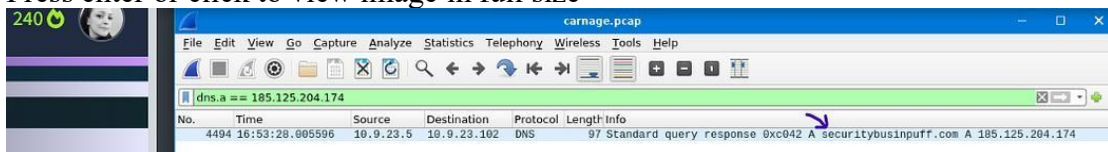


2.12. What is the **domain name of the second Cobalt Strike server IP**? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

securitybusinpu[.]com

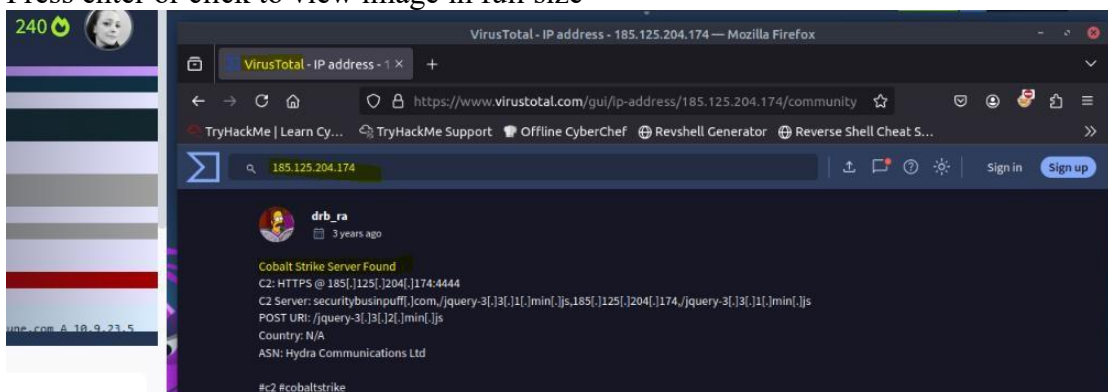
Applied filter: `dns.a == 185.125.204.174`

Press enter or click to view image in full size



Could also read the information provided in *VirtusTotal*.

Press enter or click to view image in full size



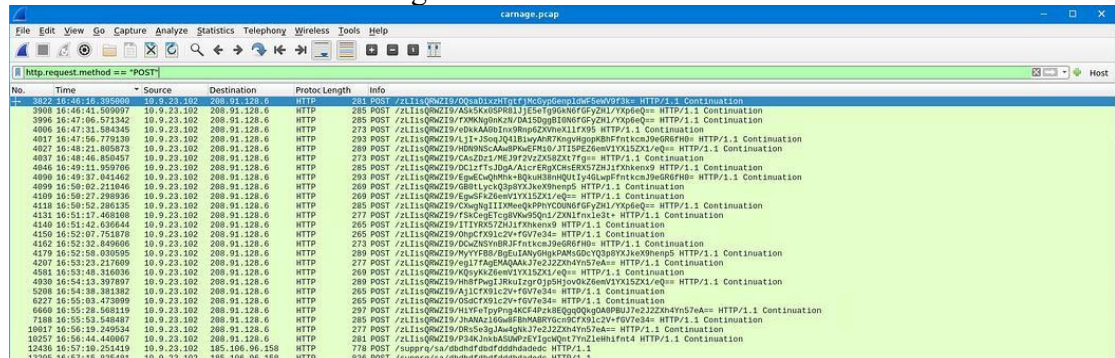
2.13. What is the **domain name of the post-infection traffic**?

maldivchost[.]net

It was provided a hint: *Filter POST HTTP traffic.*

Applied filter: *http.request.method == "POST"*.

Press enter or click to view image in full size

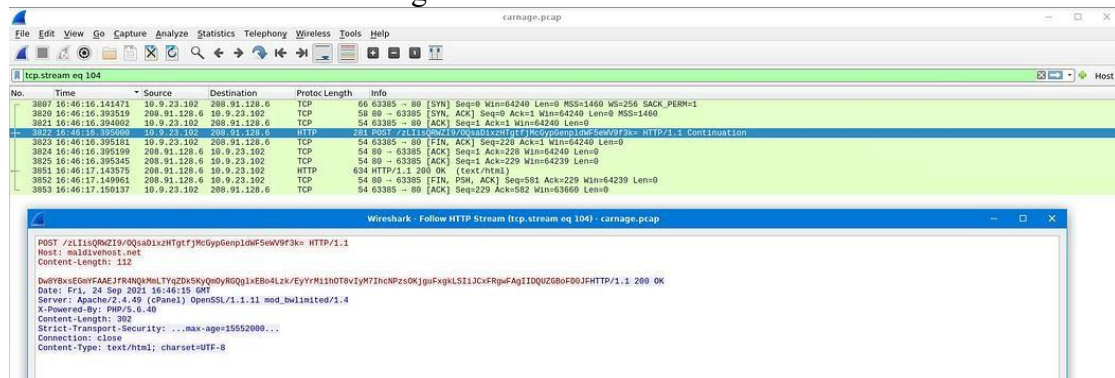


The image shows a Wireshark packet capture of an HTTP stream. The filter applied is 'http.request.method == "POST"'. The packet list shows several POST requests from 10.0.2.10 to 200.91.128.6. The selected packet (No. 281) is a POST request to /L1isQRWZ19/00a01x2HtGfFMc0yGemp1dWfSewV93k=. The packet details pane shows the request structure: HTTP/1.1, Host: maldivehost.net, Content-Length: 112, and a body containing a long alphanumeric string.

No.	Time	Source	Destination	Protoc	Length	Info
3822	10.46:10.385900	10.0.2.10	200.91.128.6	HTTP	281	POST /L1isQRWZ19/00a01x2HtGfFMc0yGemp1dWfSewV93k= HTTP/1.1 Continuation
3908	10.46:41.509997	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/ASkx0SP801jE5etg9kNfGfY2H1/vxpeq= HTTP/1.1 Continuation
3996	10.47:06.573342	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/rXNkgmKzN/DA150gB10WfGfY2H1/vxpeq= HTTP/1.1 Continuation
4006	10.47:31.584345	10.0.2.10	200.91.128.6	HTTP	273	POST /L1isQRWZ19/eBKAAB0Inx98p2ZXNeX11fX95 HTTP/1.1 Continuation
4017	10.47:56.770130	10.0.2.10	200.91.128.6	HTTP	293	POST /L1isQRWZ19/L1j-35eqQ4181yARf7Np9gpgp8dFrtckzJ9eG8f8= HTTP/1.1 Continuation
4027	10.48:21.805873	10.0.2.10	200.91.128.6	HTTP	289	POST /L1isQRWZ19/r80N8SCAdwP4eF819/715PZ26wV15ZXL1eq= HTTP/1.1 Continuation
4037	10.48:46.850457	10.0.2.10	200.91.128.6	HTTP	273	POST /L1isQRWZ19/CAz2d1/ME39fV2zX58ZK17rg= HTTP/1.1 Continuation
4045	10.49:11.959706	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/OC1fT5J0pA1cFgK8pC8u838HqU1y46LwPfrtckzJ9eG8f8= HTTP/1.1 Continuation
4098	10.49:37.841462	10.0.2.10	200.91.128.6	HTTP	293	POST /L1isQRWZ19/EgECQqMhK-BQkxH38HqU1y46LwPfrtckzJ9eG8f8= HTTP/1.1 Continuation
4099	10.50:02.211046	10.0.2.10	200.91.128.6	HTTP	269	POST /L1isQRWZ19/88B1LyCQ3p8VXkx98p5 HTTP/1.1 Continuation
4109	10.50:27.298936	10.0.2.10	200.91.128.6	HTTP	269	POST /L1isQRWZ19/EgECQqMhK-BQkxH38HqU1y46LwPfrtckzJ9eG8f8= HTTP/1.1 Continuation
4118	10.50:52.286135	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/CxngH112MeQkPPNYC0N8GfY2H1/vxpeq= HTTP/1.1 Continuation
4131	10.51:17.461068	10.0.2.10	200.91.128.6	HTTP	277	POST /L1isQRWZ19/r8kCnEtGcgVw0u051ZXL1eq= HTTP/1.1 Continuation
4140	10.51:42.636644	10.0.2.10	200.91.128.6	HTTP	265	POST /L1isQRWZ19/ITVXK57ZK1fXkx98 HTTP/1.1 Continuation
4150	10.52:07.751878	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/88B1LyCQ3p8VXkx98p5 HTTP/1.1 Continuation
4162	10.52:32.849606	10.0.2.10	200.91.128.6	HTTP	273	POST /L1isQRWZ19/OCwZCNSv8R3FrtckzJ9eG8f8= HTTP/1.1 Continuation
4179	10.52:58.030595	10.0.2.10	200.91.128.6	HTTP	289	POST /L1isQRWZ19/MyYF8B/8gU1AkyQ8pAMG0CQ3p8VXkx98p5 HTTP/1.1 Continuation
4207	10.53:23.217609	10.0.2.10	200.91.128.6	HTTP	277	POST /L1isQRWZ19/eq17fAgP4QA0A17x2J2Z2v4v57A= HTTP/1.1 Continuation
4581	10.53:48.316936	10.0.2.10	200.91.128.6	HTTP	269	POST /L1isQRWZ19/r8kCnEtGcgVw0u051ZXL1eq= HTTP/1.1 Continuation
4930	10.54:13.397897	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/88B1LyCQ3p8VXkx98p5 HTTP/1.1 Continuation
5208	10.54:38.381382	10.0.2.10	200.91.128.6	HTTP	265	POST /L1isQRWZ19/Aj1CFXK1c2V-f0V7e34= HTTP/1.1 Continuation
6227	10.55:03.473999	10.0.2.10	200.91.128.6	HTTP	265	POST /L1isQRWZ19/05CfR91c2V-f0V7e34= HTTP/1.1 Continuation
6666	10.55:28.568119	10.0.2.10	200.91.128.6	HTTP	297	POST /L1isQRWZ19/88B1LyCQ3p8VXkx98p5 HTTP/1.1 Continuation
7188	10.55:53.548487	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/JhAAZ16w8fBHMABRYC0N8GfY2H1/vxpeq= HTTP/1.1 Continuation
10017	10.56:19.249534	10.0.2.10	200.91.128.6	HTTP	277	POST /L1isQRWZ19/r8kCnEtGcgVw0u051ZXL1eq= HTTP/1.1 Continuation
10257	10.56:44.440607	10.0.2.10	200.91.128.6	HTTP	281	POST /L1isQRWZ19/P3AKJhAA50wP2Y1g0qNt7vZ1ehfnt4 HTTP/1.1 Continuation
12430	10.57:10.252419	10.0.2.10	185.106.96.158	HTTP	778	POST /supra/sa/d8dhf8d8dd8dd8 HTTP/1.1
13205	10.57:15.825481	10.0.2.10	185.106.96.158	HTTP	826	POST /supra/sa/d8dhf8d8dd8dd8 HTTP/1.1

Followed *HTTP Stream*.

Press enter or click to view image in full size



The image shows a Wireshark packet capture of an HTTP stream. The filter applied is 'http.stream eq 104'. The packet list shows several TCP and HTTP packets. The selected packet (No. 281) is a POST request to /L1isQRWZ19/00a01x2HtGfFMc0yGemp1dWfSewV93k=. The packet details pane shows the request structure: HTTP/1.1, Host: maldivehost.net, Content-Length: 112, and a body containing a long alphanumeric string.

No.	Time	Source	Destination	Protoc	Length	Info
3807	10.46:16.141471	10.0.2.10	200.91.128.6	TCP	64	63385 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM=1
3820	10.46:16.393519	200.91.128.6	10.0.2.10	TCP	58	80 → 63385 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460
3821	10.46:16.393519	10.0.2.10	200.91.128.6	TCP	54	63385 → 80 [ACK] Seq=1 Ack=229 Win=64239 Len=0
3822	10.46:16.393500	10.0.2.10	200.91.128.6	HTTP	281	POST /L1isQRWZ19/00a01x2HtGfFMc0yGemp1dWfSewV93k= HTTP/1.1 Continuation
3823	10.46:16.393519	10.0.2.10	200.91.128.6	TCP	54	63385 → 80 [FIN, ACK] Seq=229 Ack=1 Win=64240 Len=0
3824	10.46:16.393519	200.91.128.6	10.0.2.10	TCP	54	80 → 63385 [ACK] Seq=1 Ack=229 Win=64240 Len=0
3825	10.46:16.393545	200.91.128.6	10.0.2.10	TCP	54	80 → 63385 [ACK] Seq=1 Ack=229 Win=64239 Len=0
3851	10.46:17.143575	200.91.128.6	10.0.2.10	TCP	634	HTTP/1.1 200 OK (text/html)
3852	10.46:17.143961	200.91.128.6	10.0.2.10	TCP	54	80 → 63385 [FIN, PSN, ACK] Seq=581 Ack=229 Win=64239 Len=0
3853	10.46:17.156137	10.0.2.10	200.91.128.6	TCP	54	63385 → 80 [ACK] Seq=229 Ack=582 Win=3669 Len=0

Wireshark - Follow HTTP Stream (tcp.stream eq 104) - carnage.pcap

Host: maldivehost.net
Content-Length: 112

POST /L1isQRWZ19/00a01x2HtGfFMc0yGemp1dWfSewV93k= HTTP/1.1

DW8yBvE0nyFAAEJFRK9Mh1YtZDx5Kyg0yRG0g1xEBo4LzK/EyYRk110T8vYtH7InCp2s0KjgXpLS11JcXrRgAg1DQ0Z6wV15ZXL1eq= HTTP/1.1 200 OK

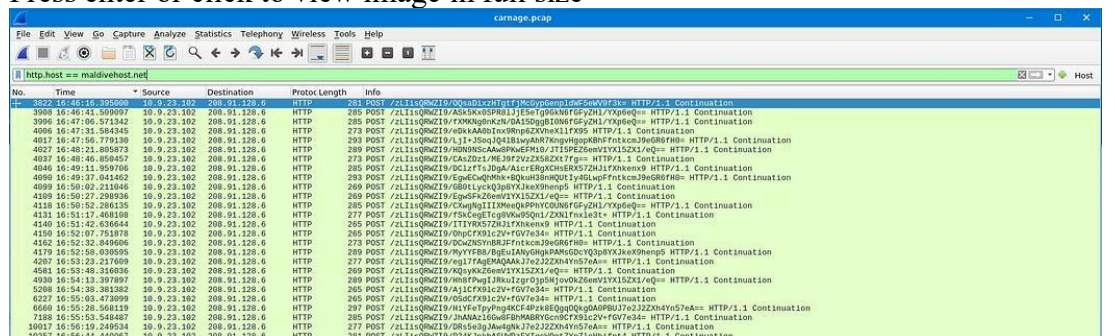
Date: Fri, 24 Sep 2022 10:46:15 GMT
Server: Apache/2.4.18 (Ubuntu) OpenSSL/1.1.1 mod_bwlimited/1.4
X-Powered-By: PHP/5.6.40
Content-Length: 382
Strict-Transport-Security: max-age=15552000...
Connection: close
Content-Type: text/html; charset=UTF-8

2.14. What are the first eleven characters that the victim host sends out to the malicious domain involved in the post-infection traffic?

L1isQRWZ19

Applied filter: *http.host == maldivehost.net*.

Press enter or click to view image in full size



The image shows a Wireshark packet capture of an HTTP stream. The filter applied is 'http.host == maldivehost.net'. The packet list shows several POST requests from 10.0.2.10 to 200.91.128.6. The selected packet (No. 281) is a POST request to /L1isQRWZ19/00a01x2HtGfFMc0yGemp1dWfSewV93k=. The packet details pane shows the request structure: HTTP/1.1, Host: maldivehost.net, Content-Length: 112, and a body containing a long alphanumeric string.

No.	Time	Source	Destination	Protoc	Length	Info
3822	10.46:10.385900	10.0.2.10	200.91.128.6	HTTP	281	POST /L1isQRWZ19/00a01x2HtGfFMc0yGemp1dWfSewV93k= HTTP/1.1 Continuation
3908	10.46:41.509997	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/ASkx0SP801jE5etg9kNfGfY2H1/vxpeq= HTTP/1.1 Continuation
3996	10.47:06.573342	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/rXNkgmKzN/DA150gB10WfGfY2H1/vxpeq= HTTP/1.1 Continuation
4006	10.47:31.584345	10.0.2.10	200.91.128.6	HTTP	273	POST /L1isQRWZ19/eBKAAB0Inx98p2ZXNeX11fX95 HTTP/1.1 Continuation
4017	10.47:56.770130	10.0.2.10	200.91.128.6	HTTP	293	POST /L1isQRWZ19/L1j-35eqQ4181yARf7Np9gpgp8dFrtckzJ9eG8f8= HTTP/1.1 Continuation
4027	10.48:21.805873	10.0.2.10	200.91.128.6	HTTP	289	POST /L1isQRWZ19/r80N8SCAdwP4eF819/715PZ26wV15ZXL1eq= HTTP/1.1 Continuation
4037	10.48:46.850457	10.0.2.10	200.91.128.6	HTTP	273	POST /L1isQRWZ19/CAz2d1/ME39fV2zX58ZK17rg= HTTP/1.1 Continuation
4045	10.49:11.959706	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/OC1fT5J0pA1cFgK8pC8u838HqU1y46LwPfrtckzJ9eG8f8= HTTP/1.1 Continuation
4098	10.49:37.841462	10.0.2.10	200.91.128.6	HTTP	293	POST /L1isQRWZ19/EgECQqMhK-BQkxH38HqU1y46LwPfrtckzJ9eG8f8= HTTP/1.1 Continuation
4099	10.50:02.211046	10.0.2.10	200.91.128.6	HTTP	269	POST /L1isQRWZ19/88B1LyCQ3p8VXkx98p5 HTTP/1.1 Continuation
4109	10.50:27.298936	10.0.2.10	200.91.128.6	HTTP	269	POST /L1isQRWZ19/EgECQqMhK-BQkxH38HqU1y46LwPfrtckzJ9eG8f8= HTTP/1.1 Continuation
4118	10.50:52.286135	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/CxngH112MeQkPPNYC0N8GfY2H1/vxpeq= HTTP/1.1 Continuation
4131	10.51:17.461068	10.0.2.10	200.91.128.6	HTTP	277	POST /L1isQRWZ19/r8kCnEtGcgVw0u051ZXL1eq= HTTP/1.1 Continuation
4140	10.51:42.636644	10.0.2.10	200.91.128.6	HTTP	265	POST /L1isQRWZ19/ITVXK57ZK1fXkx98 HTTP/1.1 Continuation
4150	10.52:07.751878	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/88B1LyCQ3p8VXkx98p5 HTTP/1.1 Continuation
4162	10.52:32.849606	10.0.2.10	200.91.128.6	HTTP	273	POST /L1isQRWZ19/OCwZCNSv8R3FrtckzJ9eG8f8= HTTP/1.1 Continuation
4179	10.52:58.030595	10.0.2.10	200.91.128.6	HTTP	289	POST /L1isQRWZ19/MyYF8B/8gU1AkyQ8pAMG0CQ3p8VXkx98p5 HTTP/1.1 Continuation
4207	10.53:23.217609	10.0.2.10	200.91.128.6	HTTP	277	POST /L1isQRWZ19/eq17fAgP4QA0A17x2J2Z2v4v57A= HTTP/1.1 Continuation
4581	10.53:48.316936	10.0.2.10	200.91.128.6	HTTP	269	POST /L1isQRWZ19/r8kCnEtGcgVw0u051ZXL1eq= HTTP/1.1 Continuation
4930	10.54:13.397897	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/88B1LyCQ3p8VXkx98p5 HTTP/1.1 Continuation
5208	10.54:38.381382	10.0.2.10	200.91.128.6	HTTP	265	POST /L1isQRWZ19/Aj1CFXK1c2V-f0V7e34= HTTP/1.1 Continuation
6227	10.55:03.473999	10.0.2.10	200.91.128.6	HTTP	265	POST /L1isQRWZ19/05CfR91c2V-f0V7e34= HTTP/1.1 Continuation
6666	10.55:28.568119	10.0.2.10	200.91.128.6	HTTP	297	POST /L1isQRWZ19/88B1LyCQ3p8VXkx98p5 HTTP/1.1 Continuation
7188	10.55:53.548487	10.0.2.10	200.91.128.6	HTTP	285	POST /L1isQRWZ19/JhAAZ16w8fBHMABRYC0N8GfY2H1/vxpeq= HTTP/1.1 Continuation
10017	10.56:19.249534	10.0.2.10	200.91.128.6	HTTP	277	POST /L1isQRWZ19/r8kCnEtGcgVw0u051ZXL1eq= HTTP/1.1 Continuation
10257	10.56:44.440607	10.0.2.10	200.91.128.6	HTTP	281	POST /L1isQRWZ19/P3AKJhAA50wP2Y1g0qNt7vZ1ehfnt4 HTTP/1.1 Continuation

Followed *HTTP Stream*.

Could have just analyzed *HTTP Stream* followed in 2.13.

Press enter or click to view image in full size



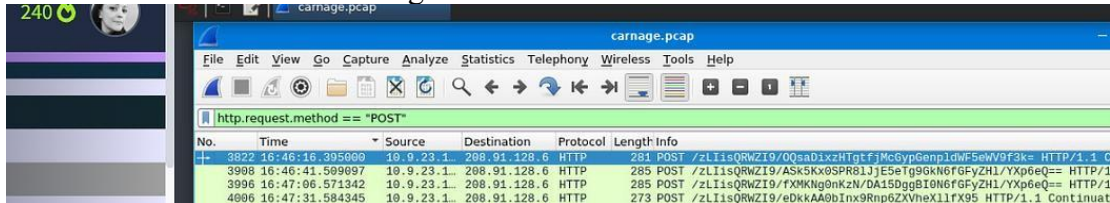
2.15. What was the length for the first packet sent out to the C2 server?

281

Applied filter: *http.request.method == "POST"*.

Looked for *Length* field.

Press enter or click to view image in full size



2.16. What was the Server header for the malicious domain from the previous question?

Apache/2.4.49 (cPanel) OpenSSL/1.1.11 mod_bwlimited/1.4

The answer is direct looking into 2.14.

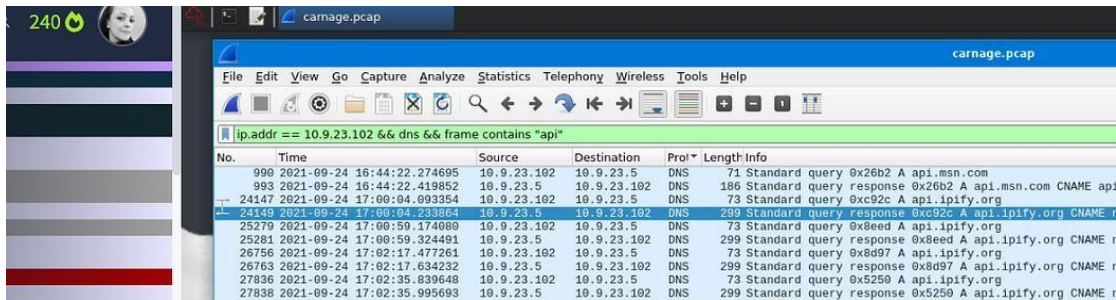
2.17. The malware used an API to check for the IP address of the victim's machine. What was the date and time when the DNS query for the IP check domain occurred? (answer format: yyyy-mm-dd hh:mm:ss UTC)

Apache/2.4.49 (cPanel) OpenSSL/1.1.11 mod_bwlimited/1.4

Our victim's machine is 10.9.23.102 .

Applied filter: *ip.addr==10.9.23.102 && dns && frame contains "api"*

Press enter or click to view image in full size

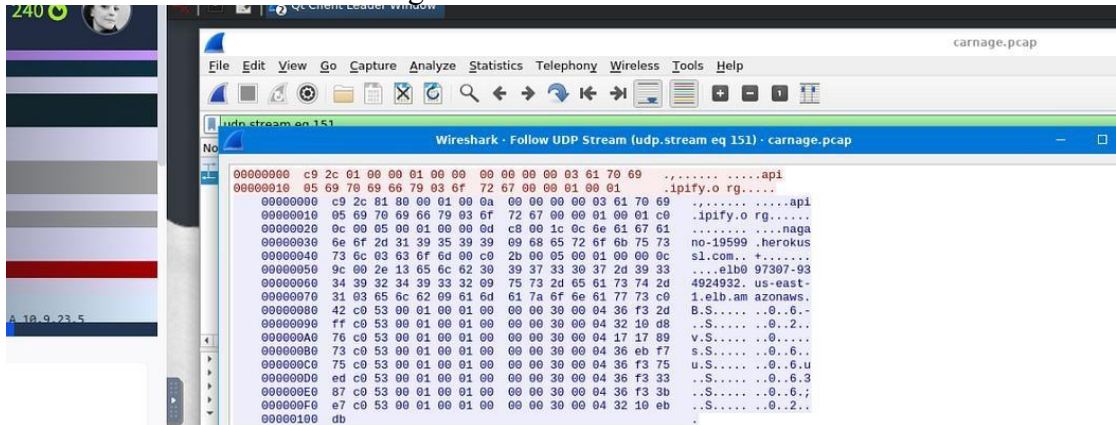


2.18. What was the **domain** in the DNS query from the previous question?

api.ipify.org

Followed *UDP Stream* related to 2.17.

Press enter or click to view image in full size



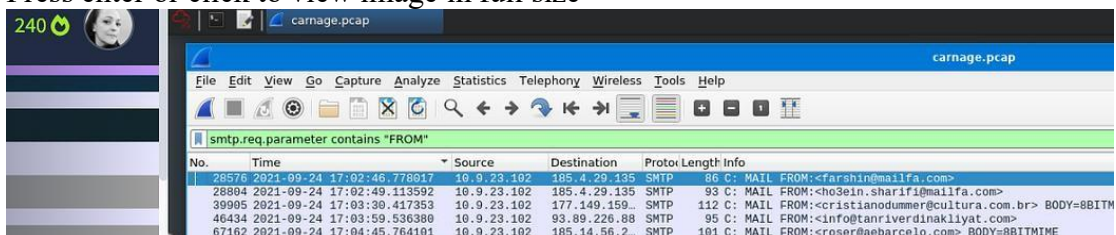
2.19. Looks like there was some malicious spam (malspam) activity going on.

What was the **first MAIL FROM** address observed in the traffic?

farshin@mailfa.com

Applied filter: *smtp.req.parameter contains "FROM"*

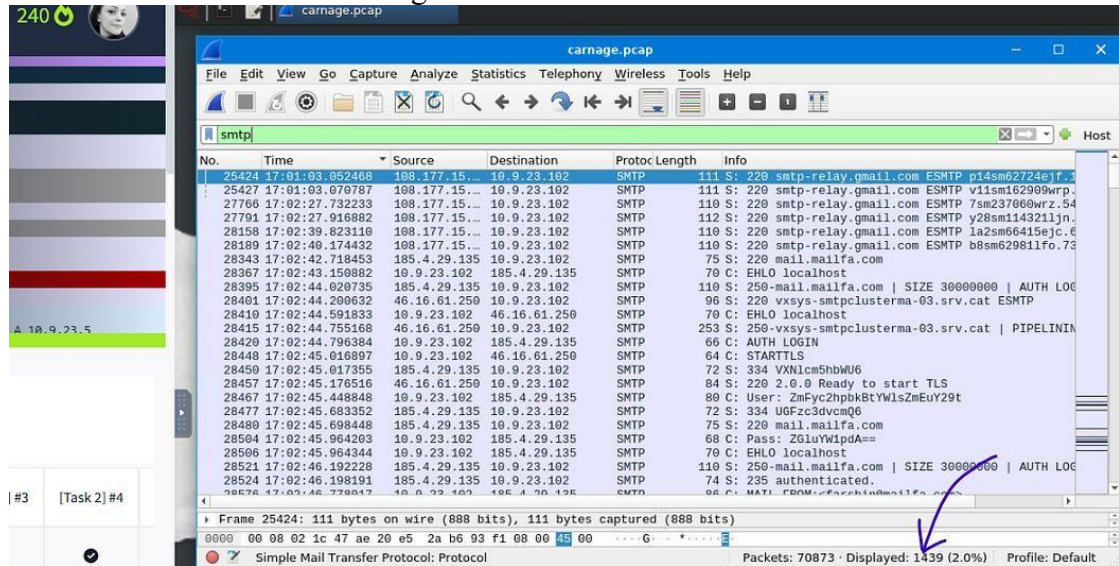
Press enter or click to view image in full size



2.20. How many packets were observed for the **SMTP** traffic?

1439

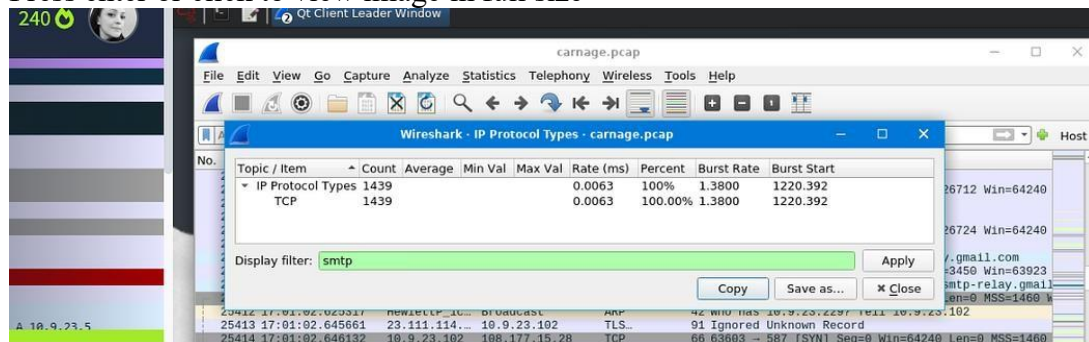
Press enter or click to view image in full size



Employed another method to find the answer which was:

- Clicked *Statistics* in the main menu bar.
- Clicked *IPv4 Statistics*.
- Clicked *IP Protocol Types*.
- Typed *smtp* in *Display filter*: field.
- Clicked *Apply*.

Press enter or click to view image in full size





Result:

Successfully identified malicious domains, downloaded payloads, Cobalt Strike C2 servers, post-infection traffic, SMTP malspam activity, and indicators of compromise (IOCs).